

MODULE 06

# Wrist & Hand MRI: Ligaments, Tendons & Classification Systems

TFCC Palmer · SL Geissler · Kienbock Lichtman · Stener · ECU · Flexor Pulleys · CMC OA · Mayfield

1.0 AMA PRA Cat 1

Wrist/Hand

CME Article Review

## LEARNING OBJECTIVES

1. Apply the Palmer classification to TFCC tears (Types 1A–1D traumatic, 2A–2E degenerative).
2. Grade scapholunate ligament injuries using the Geissler classification and recognize the SL gap/Terry Thomas sign.
3. Recognize and classify thumb UCL injuries, including the Stener lesion and its surgical implications.
4. Apply the Lichtman classification to stage Kienbock disease.
5. Classify first CMC osteoarthritis using the Eaton-Littler system.
6. Grade flexor pulley injuries and quantify bowstringing on axial MRI.
7. Apply the Mayfield classification to perilunate instability and recognize lunate dislocation on radiograph.

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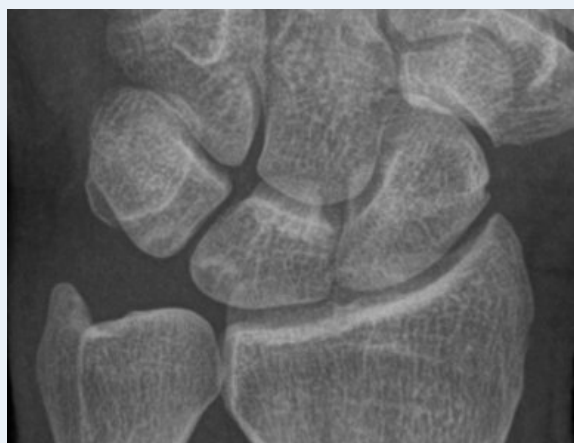
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## 1. TFCC Injuries — Palmer Classification

The triangular fibrocartilage complex (TFCC) is the primary stabilizer of the distal radioulnar joint (DRUJ) and the ulnar wrist. The Palmer classification divides TFCC tears into traumatic (Type 1, subtypes A–D) and degenerative (Type 2, subtypes A–E) categories, and the distinction matters directly for management: traumatic peripheral tears are often repairable, while degenerative tears are addressed by treating the underlying ulnar impaction rather than direct TFCC repair.

### Palmer Classification — TFCC Tears

| Type             | Subtype               | Description                                                                | Repairability                                         |
|------------------|-----------------------|----------------------------------------------------------------------------|-------------------------------------------------------|
| 1 (Traumatic)    | 1A — Central disc     | Horizontal perforation of the central articular disc                       | Non-repairable; debridement                           |
| 1 (Traumatic)    | 1B — Peripheral ulnar | Detachment at the ulnar styloid/fovea                                      | Best repair candidate; arthroscopic suture            |
| 1 (Traumatic)    | 1C — Distal           | Tear of the ulnocarpal ligaments                                           | Repair if DRUJ unstable                               |
| 1 (Traumatic)    | 1D — Radial           | Avulsion from the sigmoid notch of the radius                              | Repair; associated DRUJ instability                   |
| 2 (Degenerative) | 2A–2E                 | Progressive chondromalacia → perforation → lunotriquetral tear → arthrosis | Ulnar shortening osteotomy if positive ulnar variance |



*Positive ulnar variance on PA wrist radiograph — a key risk factor for degenerative (Type 2) TFCC tears and ulnocarpal impaction.*

- Type 1B (peripheral ulnar) tears detach at the fovea and are the most surgically important subtype — they are the best candidates for arthroscopic repair. Degenerative Type 2 tears are instead managed by treating the positive ulnar variance itself, typically with ulnar shortening osteotomy.

### Key References

- Palmer AK. Triangular fibrocartilage complex lesions: a classification. J Hand Surg Am 1989.

## 2. Scapholunate Ligament Injury — Geissler Grading

Scapholunate (SL) ligament injury is the most common cause of carpal instability and, if missed, progresses predictably to scapholunate advanced collapse (SLAC) wrist. The Geissler classification, originally an arthroscopic grading system, correlates closely with MRI findings and carpal alignment measurements, making it a useful shared framework between radiologist and hand surgeon.

### Geissler Classification — Arthroscopic / MRI Correlation

| Grade | MRI Finding                                                    | Arthroscopic Correlate                                     | Instability                  |
|-------|----------------------------------------------------------------|------------------------------------------------------------|------------------------------|
| I     | Ligament intact or minor signal change; no carpal malalignment | Attenuation/hemorrhage; no probe passage                   | None                         |
| II    | Partial SL tear; no carpal malalignment                        | Incongruity; probe palpation confirms partial tear         | None or minimal              |
| III   | Partial or complete tear; SL gap present                       | Probe (not arthroscope) passes between scaphoid and lunate | Step-off; ligamentous laxity |
| IV    | Complete tear; DISI alignment                                  | Arthroscope (2.7mm) passes freely through the gap          | Gross instability; DISI      |

### Key SL Measurements

| Measurement                   | Normal | Abnormal                                                                             |
|-------------------------------|--------|--------------------------------------------------------------------------------------|
| SL gap (PA radiograph)        | < 3mm  | > 3mm — Terry Thomas sign (SL ligament disruption); > 5mm — gross dissociation       |
| Scapholunate angle (sagittal) | 30–60° | > 70° → DISI pattern (dorsal intercalated segment instability); < 30° → VISI pattern |



PA wrist radiograph demonstrating normal carpal alignment, used as a baseline reference for assessing SL gap.

- The SL gap measurement is named the "Terry Thomas sign" after the British comedian known for his distinctive gap-toothed smile — a gap greater than 3mm on a PA wrist radiograph should prompt close evaluation of the scapholunate ligament on MRI, even when not the primary clinical question.

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### Key References

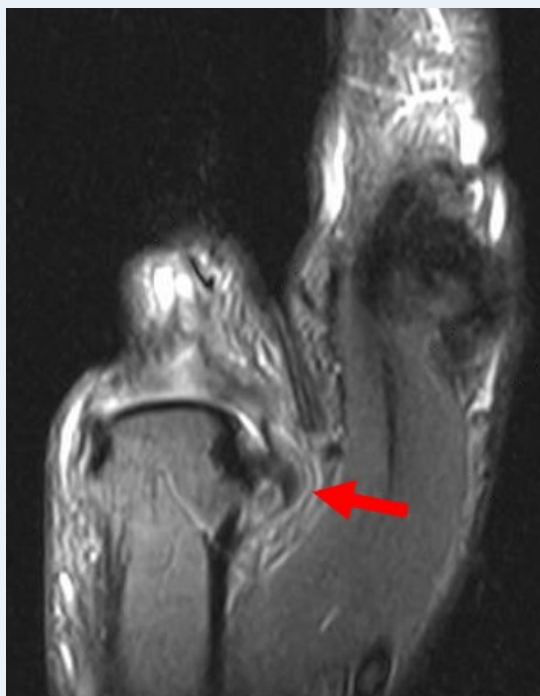
- Linscheid RL, Dobyns JH, Beabout JW, Bryan RS. Traumatic instability of the wrist. J Bone Joint Surg Am 1972.
- Ringler MD. MRI of wrist ligaments. J Hand Surg Am 2013.

### 3. Thumb UCL Injury & the Stener Lesion

Ulnar collateral ligament (UCL) injury of the thumb metacarpophalangeal joint — classically "skier's thumb" — ranges from a simple sprain to complete rupture. The single most important distinction for management is whether the torn ligament has become a Stener lesion: when the proximal UCL stump retracts and folds back superficial to the adductor aponeurosis, it can never heal in apposition without surgery, regardless of how the joint is immobilized.

#### Thumb UCL Injury Grading

| Grade                           | Finding                                                                                                                                                        |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I (Sprain)                      | Ligament thickened/edematous; fibers continuous; no instability                                                                                                |
| II (Partial tear)               | Partial fiber discontinuity; ligament remains in anatomic position relative to the adductor aponeurosis                                                        |
| III (Complete tear, non-Stener) | Complete discontinuity; torn end remains deep to the adductor aponeurosis — may heal with immobilization                                                       |
| Stener lesion                   | Torn proximal UCL stump displaced superficial to (folded over) the adductor aponeurosis — the classic "yo-yo on a string" appearance; requires surgical repair |



*Coronal MRI of a Stener lesion: the torn UCL stump is displaced superficial to the adductor aponeurosis.*

! Coronal T1 or PD fat-saturated images at the level of the first MCP joint are the optimal sequence and plane for identifying a Stener lesion. Missing this distinction has real consequences: a Grade III tear without Stener may be treated conservatively, while a Stener lesion requires surgery to restore apposition of the torn ends.

#### Key References

- Stener B. Displacement of the ruptured ulnar collateral ligament of the metacarpophalangeal joint of the thumb. J Bone Joint Surg Br 1962.
- Hergan K, Mittler C, Oser W. Sonography and MRI of the thumb ulnar collateral ligament: a comparative study. Eur Radiol 1997.

## 4. Kienbock Disease — Lichtman Classification

Kienbock disease is avascular necrosis of the lunate, classically associated with negative ulnar variance, which is thought to increase shear loading on the lunate. The Lichtman classification stages disease progression from isolated marrow signal change through collapse and ultimately pancarpal arthrosis, and the distinction between Stage IIIA and IIIB is the key surgical branch point.



*Kienbock disease: radiograph and coronal MRI showing lunate involvement, with corresponding sclerosis and signal change.*

### Lichtman Classification — Kienbock Disease

| Stage | Findings                                                                                                   |
|-------|------------------------------------------------------------------------------------------------------------|
| I     | Normal radiograph; MRI shows diffuse lunate marrow signal change (low T1, variable T2)                     |
| II    | Radiographic sclerosis of the lunate; no collapse                                                          |
| IIIA  | Lunate collapse WITHOUT fixed scaphoid rotation                                                            |
| IIIB  | Lunate collapse WITH fixed scaphoid rotation (DISI deformity) — critical distinction for surgical planning |
| IV    | Pancarpal arthrosis (end-stage)                                                                            |

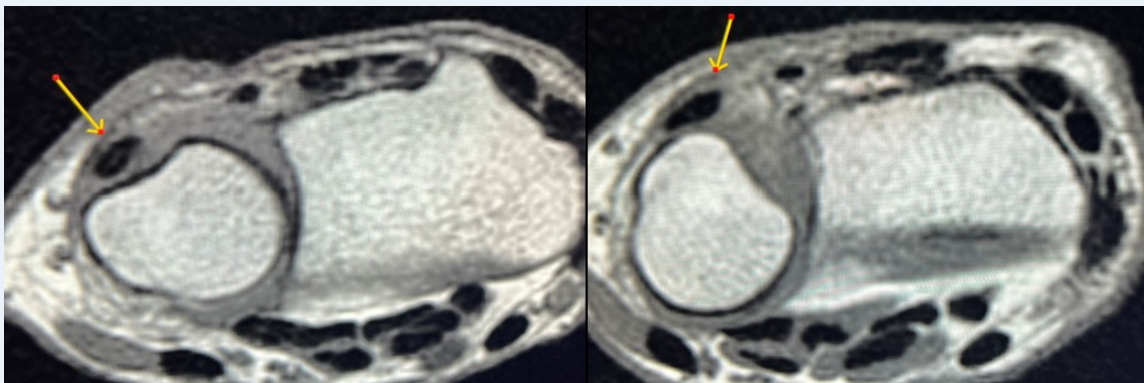
! The IIIA versus IIIB distinction directly changes surgical strategy: IIIA may be managed with joint-preserving procedures (e.g. radial shortening osteotomy), while IIIB — with fixed scaphoid rotation — typically requires more extensive reconstruction or salvage procedures.

### Key References

- Lichtman DM, Mack GR, MacDonald RI, et al. Kienbock's disease: the role of silicone replacement arthroplasty. J Hand Surg Am 1977.

## 5. ECU Tendon Instability

Extensor carpi ulnaris (ECU) tendon subluxation or dislocation from its groove in the distal ulna is a common cause of ulnar-sided wrist pain in racquet sport athletes, resulting from tearing of the ECU subsheath. Because subluxation is position-dependent, imaging in a single static position can miss it entirely — dynamic or comparative positioning is often necessary.



*Axial MRI through the ulnar styloid in two positions, used to assess ECU tendon position and subsheath integrity for subluxation.*

- Axial PD fat-saturated images through the ulnar styloid, ideally obtained in both supination and pronation for comparison, are the key sequence and technique for evaluating ECU subluxation — imaging in neutral position alone risks a false-negative study.

## 6. First CMC Osteoarthritis & Flexor Pulley Injuries

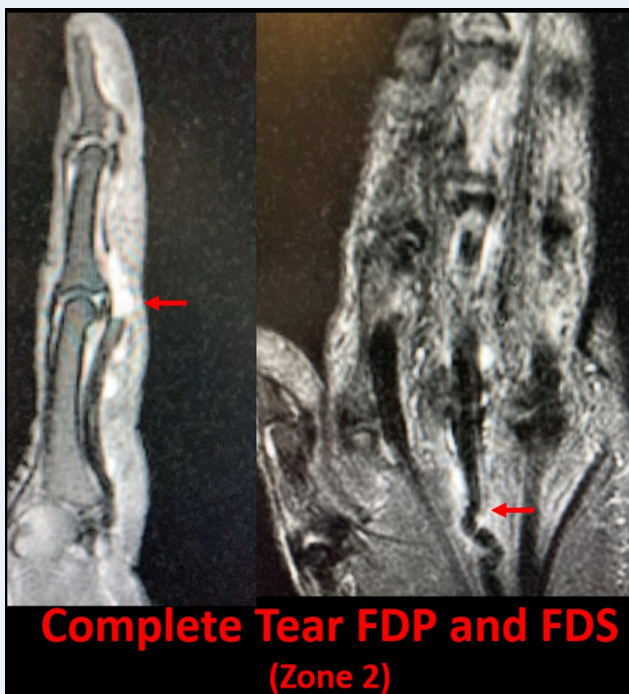
First carpometacarpal (CMC) osteoarthritis is among the most common sites of hand osteoarthritis and is staged with the Eaton-Littler system based on joint space narrowing and the degree of subluxation. Separately, flexor pulley injuries — classically a rock-climbing injury — are graded by the degree of tendon "bowstringing" away from the underlying bone on flexion-position imaging.

### Eaton-Littler Classification — First CMC Osteoarthritis

| Stage | Radiographic Findings                                                                           |
|-------|-------------------------------------------------------------------------------------------------|
| I     | Normal joint space ( $\leq 1$ mm narrowing); no subchondral change                              |
| II    | Joint space narrowing with sclerosis and osteophytes; subluxation $< 1/3$ of joint width        |
| III   | Subluxation $> 1/3$ of joint width; larger osteophytes ( $> 2$ mm); no pantrapezial involvement |
| IV    | Pantrapezial arthrosis — both the CMC and scaphotrapezotrapezoid (STT) joints involved          |

### Flexor Pulley Injury Grading (A2/A4) — Axial MRI

| Grade                | Finding                                                    | Bowstringing                                       |
|----------------------|------------------------------------------------------------|----------------------------------------------------|
| 1 — Sprain           | Pulley thickening; fibers intact                           | None ( $< 1$ mm)                                   |
| 2 — Partial tear     | Partial fiber disruption; attenuated pulley                | $< 3$ mm (partial bowstringing)                    |
| 3 — Complete rupture | Full-thickness disruption of a single pulley (commonly A2) | $> 3$ mm — pathognomonic for complete rupture      |
| 4 — Multiple pulleys | Combined A2 + A3 or A2 + A4 complete tears                 | Severe; typically requires surgical reconstruction |



*Sagittal and coronal MRI of a complete flexor digitorum profundus and superficialis tear in Zone 2, illustrating the severe end of the flexor tendon/pulley injury spectrum.*

- Bowstringing is measured as the distance between the flexor tendon and the volar cortex of the proximal phalanx on axial MRI with the finger in flexion. A measurement greater than 3mm is essentially pathognomonic for complete pulley rupture and should be stated explicitly in the report, since it directly determines whether surgical reconstruction is needed.

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### Key References

- Eaton RG, Littler JW. Ligament reconstruction for the painful thumb carpometacarpal joint. J Bone Joint Surg Am 1973.
- Klauser A, Frauscher F, Bodner G, et al. Finger pulley injuries in extreme rock climbers: depiction with dynamic ultrasound. Radiology 2002.

## 7. Perilunate Instability — Mayfield Classification

Perilunate instability results from a progressive ligamentous injury pattern around the lunate, typically from a fall on an outstretched hand with the wrist in extension. The Mayfield classification describes four stages of progressive carpal ligament disruption, and the key distinction for recognition on imaging is between perilunate dislocation (the lunate remains aligned with the radius, while the rest of the carpus dislocates dorsally) and frank lunate dislocation (the lunate itself rotates and displaces volarly, producing the classic "spilled teacup" appearance).

### Mayfield Classification — Progressive Perilunate Instability

| Stage | Ligamentous Injury                                    | Imaging Pattern                                                                                                              |
|-------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| I     | Scapholunate ligament disruption                      | SL widening (Terry Thomas sign); no dislocation                                                                              |
| II    | Stage I plus capitulunate disruption                  | Perilunate dislocation begins; capitate displaced dorsal to lunate                                                           |
| III   | Stage II plus lunotriquetral ligament disruption      | Complete perilunate dislocation; the lunate remains aligned with the radius while the rest of the carpus dislocates dorsally |
| IV    | Stage III plus dorsal radiocarpal ligament disruption | Frank lunate dislocation — the lunate itself rotates and displaces volarly ("spilled teacup" sign)                           |



*Lateral wrist radiograph demonstrating lunate dislocation — the lunate is rotated and displaced volarly out of its normal articulation with the radius, the "spilled teacup" appearance of Mayfield Stage IV injury.*

! The distinction between perilunate dislocation (Stage III) and lunate dislocation (Stage IV) is easy to miss quickly on a single AP radiograph — the lateral view is essential, since it directly shows whether the lunate remains aligned with the distal radius or has rotated volarly out of its normal articulation. Both represent surgical emergencies requiring urgent reduction to prevent median nerve injury and avascular necrosis of the lunate.

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### Key References

- Mayfield JK, Johnson RP, Kilcoyne RK. Carpal dislocations: pathomechanics and progressive perilunar instability. J Hand Surg Am 1980.